
Education

2017 - 2021

Massachusetts Institute of Technology (MIT), Cambridge, MA

B.S. in Computer Science and Engineering, B.S. in Mathematics, Minor in Economics

GPA: 5.0/5.0

Relevant Coursework: Machine Learning, Computer Vision, Computational Cognitive Science, Cryptography, Algorithms, Information Theory, Statistics, Real Analysis

Experience

Jun 2021 - Present

Google, Security & Anti-Abuse Research Team

Senior Software Engineer

- o Building adversarially robust, multilingual, and efficient NLP models using novel metric learning techniques, such as RETSim [1] and RETVec [3]. Deployed large-scale security and anti-abuse protections based on our research in multiple Google products (Gmail, YouTube, Workspace), e.g. blocking billions of spam emails in Gmail daily.
- o Investigating novel applications of ML for security, such as LLMs for code vulnerability detection, AI-powered file-type detection [2], and side-channel attacks on cryptographic hardware [4].
- o Conducting research on the security and privacy of LLMs, including on-device text safety protections (Google I/O 2023 demo and launched for Android), as well as data-centric approaches to reduce privacy risk and improve generalization capabilities.

Sep 2020 - Jun 2021

University of Pennsylvania, Department of Neuroscience

Research Assistant, advised by Prof. Wenqin Luo

- o Designed the convolutional recurrent neural network (CRNN) used for Scratch-AID, a deep learning-based tool which can automatically identify and quantify mouse scratching behavioral patterns from raw video footage [6].
- o Investigated the neurobiological relationship between behavioral states and breathing patterns using clustering techniques; built a classifier capable of distinguishing between 9 different behavioral states in rodents using breathing recordings [5].

Jun 2020 - Aug 2020

Google, Security & Anti-Abuse Research Team

Software Engineer Intern, hosted by Elie Bursztein

- o Built a deep-learning model training framework for security research, which supported state-of-the-art model training techniques including hyperparameter tuning, semi-supervised labeling, and transfer learning.
- o Designed and wrote a TensorFlow/Keras package which automates searching for and applying the best data augmentation policies during model training.

Sep 2019 - Dec 2019

MIT CSAIL, Medical Vision Group

Research Assistant, advised by Prof. Polina Golland

- o Built an ML model for automatically quantifying the severity of pulmonary edema from patients' x-ray images and radiology reports, helping to improve clinicians' abilities to provide more accurate and personalized treatment plans for heart failure patients.

Jun 2019 - Aug 2019

Microsoft, Edge Browser Experiences Team

Software Engineer Intern

- o Designed and shipped the quick-access Favorites toolbar button and drop-down menu in Microsoft Edge.

Papers

- [1] **Marina Zhang**, Owen Vallis, Aysegul Bumin, Tanay Vakharia, and Elie Bursztein. "RETSim: Resilient and Efficient Text Similarity." In *International Conference on Learning Representations (ICLR)*, 2024. [\[Link\]](#)
- [2] Yanick Frantantonio, Luca Invernizzi, Loua Farah, Kurt Thomas, **Marina Zhang**, ..., Elie Bursztein. "Magika: AI-Powered Content Type Detection." Under review at *International Conference on Software Engineering (ICSE) 2025*. [\[Link\]](#)
- [3] Elie Bursztein, **Marina Zhang**, Owen Vallis, Xinyu Jia, and Alexey Kurakin. "RETVec: Resilient and Efficient Text Vectorizer." In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. [\[Link\]](#)
- [4] Elie Bursztein, Luca Invernizzi, Karel Král, Daniel Moghimi, Jean-Michel Picod, and **Marina Zhang**. "Generic Attacks against Cryptographic Hardware through Long-Range Deep Learning." In *Conference on Cryptographic Hardware and Embedded Systems (CHES)*, 2024. [\[Link\]](#) (*alphabetical authorship*)
- [5] Emma Janke, **Marina Zhang**, Sang Eun Ryu, Janardhan Bhattarai, Mary Schreck, Andrew Moberly, Wenqin Luo, Long Ding, Daniel Wesson, and Minghong Ma. "Machine Learning-based Clustering and Classification of Mouse Behaviors via Respiratory Patterns." In *iScience (Cell Press) Vol. 25*, 2022. [\[Link\]](#)
- [6] Huasheng Yu, et al. "Scratch-AID: A Deep-learning Based System for Automatic Detection of Mouse Scratching Behavior with High Accuracy." In *eLife Vol. 11*, 2022. [\[Link\]](#)

Open-Source Projects

- 2023 - Present **Magika: AI-Powered Content-Type Detection**, *Contributor*
Package for efficiently and accurately detecting file content types with deep learning [2]. 7k+ stars on GitHub.
- 2023 - Present **UniSim: Universal Similarity**, *Main Contributor*
UniSim is a Python package for efficient similarity computation, fuzzy matching, and data clustering using similarity embeddings [1].
- 2022 - Present **RETVec: Resilient and Efficient Text Vectorizer**, *Main Contributor*
The RETVec package offers an easy-to-use TensorFlow/Keras API for RETVec, an efficient, multilingual, and robust text vectorizer from our research [3].
- 2021 - Present **TensorFlow Similarity: Metric Learning for Humans**, *Contributor*
TensorFlow Similarity is a TensorFlow library for similarity learning, including algorithms and models for self-supervised learning, metric learning, and contrastive learning.

Awards & Honors

- 2024 NSF GRFP Fellowship Recipient
- 2020 Tau Beta Pi (TBP) Honor Society
- 2020 IEEE Eta Kappa Nu (IEEE-HKN) Honor Society
- 2020 MIT EECS Undergraduate Research and Innovation Scholar
- 2018, 2019 2x ITA Scholar-Athlete Award
- 2018, 2019 2x NCAA Championship Elite Eight (MIT Women's Tennis)
- 2017 National Merit Scholarship Winner
- 2017 National AP Scholar

Activities

- 2022 - 2024 **Google Intern Host**
Hosted/co-hosted two Research Scientist interns and two SWE interns at Google
- 2020 - 2021 **MIT IEEE-HKN Tutor for EECS**
Tutor for 6.009 Fundamentals of Programming and 6.006 Introduction to Algorithms

2019 - 2020 **MIT xFair Committee**

Organizational committee for xFair, MIT's largest student-run career fair and tech expo

2019 - 2020 **MIT UA Innovation Committee**

Worked on projects to improve student life, study spaces, and mental health at MIT

2017 - 2019 **MIT Varsity Tennis Team**

2x NCAA Elite 8; 2x ITA Scholar-Athlete; 2x NEWMAC First Team All-Conference